You must include in your presentation:

- Who the mathematician/philosopher is. Did they have unrelated occupations.or interests?
- What they contributed to the development of Calculus.
- When their contributions were made.
- Where were they during this timeline? Were they studying or working with other mathematicians/philosophers at the time?
- Why - what led them to these discoveries? Was it in an area they were studying by choice, or did they stumble upon it while working on something else?
- Include any interesting information about the mathematician, this may include developments they made in other branches of mathematics or it just may be "Fun Fact".

You MUST submit a bibliography using APA format for your citations. (This will be handed in as a hard copy).

https://www.britannica.com/biography/Leonhard-Euler

Leonhard Euler

- Swiss mathematician and physicist
- Contributions to geometry, calculus, mechanics, number theory, astronomy, applied mathematics
- Wrote *Institutiones calculi differentialis* in 1755 and *Institutiones calculi integralis* in 1768–70
    - These two textbooks served as “prototypes to the present,” containing formulae for differentiation and indefinite integration (many invented by himself)
- Introduced notation
    - “such as Σ for the sum; the symbol *e* for the base of natural logarithms; *a*, *b* and *c* for the sides of a triangle and A, B, and C for the opposite angles; the letter *f* and parentheses for a function; and *i* for Square root of$\sqrt{-1}$. He also popularized the use of the symbol π (devised by British mathematician William Jones) for the ratio of circumference to diameter in a circle.”
- Blind in 1 eye by 1735, fully blind by 1766
- helped to establish mathematical education in Russia
- Succeeded by Lagrange
- Timeline
    - born April 15, 1707, Basel, Switzerland
    - In 1727 he moved to St. Petersburg, where he became an associate of the St. Petersburg Academy of Science
    - in 1733 succeeded Daniel Bernoulli (son of Johann Bernoulli, one of the first mathematicians in Europe) to the chair of mathematics
    - in 1735 lost the sight of one eye
    - invited by Frederick the Great in 1741, he became a member of the Berlin Academy
        - where for 25 years he produced a steady stream of publications, many of which he contributed to the St. Petersburg Academy, which granted him a pension.

- In 1748, in his *Introductio in analysin infinitorum*, he developed the concept of function in mathematical analysis
- Euler's textbooks in calculus, *Institutiones calculi differentialis* in 1755 and *Institutiones calculi integralis* in 1768–70, have served as prototypes to the present because they contain formulas of differentiation and numerous methods of indefinite integration, many of which he invented himself, for determining the work done by a force and for solving geometric problems, and he made advances in the theory of linear differential equations, which are useful in solving problems in physics
- After Frederick the Great became less cordial toward him, Euler in 1766 accepted the invitation of Catherine II to return to Russia. Soon after his arrival at St. Petersburg, a cataract formed in his remaining good eye, and he spent the last years of his life in total blindness
- is *Lettres à une princesse d'Allemagne* in 1768–72 were an admirably clear exposition of the basic principles of mechanics, optics, acoustics, and physical astronomy.
- Throughout his life Euler was much absorbed by problems dealing with the theory of numbers, which treats of the properties and relationships of integers, or whole numbers (0, ±1, ±2, etc.); in this, his greatest discovery, in 1783, was the law of quadratic reciprocity, which has become an essential part of modern number theory.
- September 18, 1783, St. Petersburg, Russia

https://mathshistory.st-andrews.ac.uk/Biographies/Euler/

- ... after 1730 he carried out state projects dealing with cartography, science education, magnetism, fire engines, machines, and ship building. ... The core of his research program was now set in place: number theory; infinitary analysis including its emerging branches, differential equations and the calculus of variations; and rational mechanics. He viewed these three fields as intimately interconnected. Studies of number theory were vital to the foundations of calculus, and special functions and differential equations were essential to rational mechanics, which supplied concrete problems.
- During the twenty-five years spent in Berlin, Euler wrote around 380 articles. He wrote books on the calculus of variations; on the calculation of planetary orbits; on artillery and ballistics (extending the book by Robins); on analysis; on shipbuilding and navigation; on the motion of the moon; lectures on the differential calculus; and a popular scientific publication *Letters to a Princess of Germany* (3 vols., 1768-72).
- He made decisive and formative contributions to geometry, calculus and number theory. He integrated Leibniz's differential calculus and Newton's method of fluxions into mathematical analysis. He introduced beta and gamma functions, and integrating factors for differential equations. He studied continuum mechanics, lunar theory with Clairaut, the three body problem, elasticity, acoustics, the wave theory of light, hydraulics, and music. He laid the foundation of analytical mechanics, especially in his *Theory of the Motions of Rigid Bodies* (1765).

- We owe to Euler the notation f(x) for a function (1734), e for the base of natural logs (1727),ii for the square root of -1 (1777), π for pi, ∑ for summation (1755), the notation for finite differences Δy and $\Delta^2 y$ and many others.

One could claim that mathematical analysis began with Euler. In 1748 in *Introductio in analysin infinitorum* Euler made ideas of Johann Bernoulli more precise in defining a function, and he stated that mathematical analysis was the study of functions. This work bases the calculus on the theory of elementary functions rather than on geometric curves, as had been done previously. Also in this work Euler gave the formula

$$e^{ix} = \cos x + i \sin x.$$

In *Introductio in analysin infinitorum* Euler dealt with logarithms of a variable taking only positive values although he had discovered the formula

$$\ln(-1) = \pi i$$

in 1727. He published his full theory of logarithms of complex numbers in 1751.

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In 1755 Euler published *Institutiones calculi differentialis* which begins with a study of the calculus of finite differences. The work makes a thorough investigation of how differentiation behaves under substitutions.

In *Institutiones calculi integralis* (1768-70) Euler made a thorough investigation of integrals which can be expressed in terms of elementary functions. He also studied beta and gamma functions, which he had introduced first in 1729. Legendre called these 'Eulerian integrals of the first and second kind' respectively while they were given the names beta function and gamma function by Binet and Gauss respectively. As well as investigating double integrals, Euler considered ordinary and partial differential equations in this work.

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The calculus of variations is another area in which Euler made fundamental discoveries. His work *Methodus inveniendi lineas curvas* ... published in 1740 began the proper study of the calculus of variations. In [12] it is noted that Carathéodory considered this as:-

> *... one of the most beautiful mathematical works ever written.*

Problems in mathematical physics had led Euler to a wide study of differential equations. He considered linear equations with constant coefficients, second order differential equations with variable coefficients, power series solutions of differential equations, a method of variation of constants, integrating factors, a method of approximating solutions, and many others. When considering vibrating membranes, Euler was led to the Bessel equation which he solved by introducing Bessel functions.

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In *Mechanica* Euler considered the motion of a point mass both in a vacuum and in a resisting medium. He analysed the motion of a point mass under a central force and also considered the motion of a point mass on a surface. In this latter topic he had to solve various problems of differential geometry and geodesics.

Mechanica was followed by another important work in rational mechanics, this time Euler's two volume work on naval science. It is described in [24] as:-

> *Outstanding in both theoretical and applied mechanics, it addresses Euler's intense occupation with the problem of ship propulsion. It applies variational principles to determine the optimal ship design and first established the principles of hydrostatics ... Euler here also begins developing the kinematics and dynamics of rigid bodies, introducing in part the differential equations for their motion.*

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https://www.oiirj.org/oiirj/may-june2016/08.pdf

Euler was the first who gave f(x) as the notation to a function of x. In his book, “Introduction to the analysis of the infinite” he defined the exponential and logarithmic functions as limits,

$$e^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n$$

$$\ln x = \lim_{n\to\infty} n(x^{\frac{1}{n}} - 1)$$

By giving x the value one in the exponential series gives the Euler number e = 2.71828183…. In 1740 he discovered the universal Euler constant γ given by

$$\gamma = \lim_{n\to\infty}(1+\frac{1}{2}+\frac{1}{3}+\text{----------}+\frac{1}{n} - \ln n)=0.577215665\text{----}$$

which gives the amazing relation between logarithmic function and harmonic series

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EULER AND LAGRANGE

Euler and Lagrange worked on calculus of variation of which the main problem is to determine the function which extremises the functional and they proved that the necessary and sufficient condition for extremisation is to satisfy the non linear second order differential equation $\frac{\partial F}{\partial x} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$ which is the well known Euler-Lagrange equation. Euler generalized this problem for n dependent variables.He also considered the same problem with the help constraint called isoperimetric problems.

Euler devoted to calculus by deriving many formulas for the derivative of the rational, trigonometric and transcendental functions. He made many significant contributions to the theory of ordinary differential equations and introduced the concept of integrating factor for solving linear ordinary differential equation with constant coefficient. For partial differential equation he derives homogeneous functions and proved that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial x} = nu$ known as Euler’s theorem for homogeneous functions of degree n and generalized the result for m variables.

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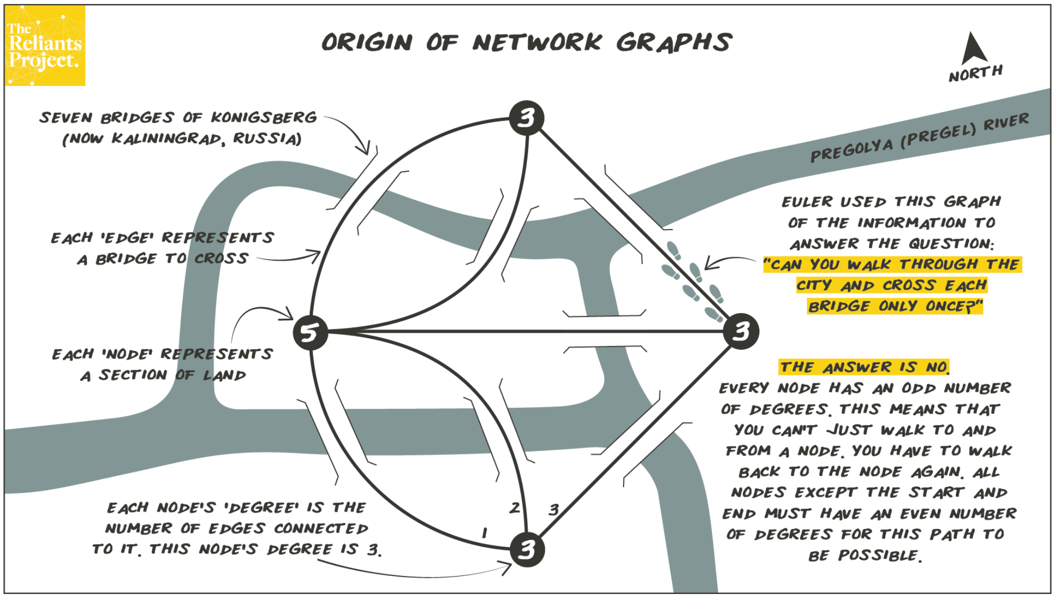
Other things to add

- Johann’s Bernoulli’s influence on Euler
    - https://old.maa.org/press/periodicals/convergence/euler-and-the-bernoullis-learning-by-teaching-introduction
    - “*I soon found an opportunity to be introduced to a famous professor Johann Bernoulli... True, he was very busy and so refused flatly to give me private lessons; but he gave me much more valuable advice to start reading more difficult mathematical books on my own and to study them as diligently as I could; if I came across some obstacle or difficulty, I was given permission to visit him freely every Sunday afternoon and he kindly explained to me everything I could not understand … and this, undoubtedly, is the best method to succeed in mathematical subjects. - Leonard Euler [13, p. 90]*”
    - Also more stories at the bottom
- Fun facts about Euler???
    - Seven Bridges of Königsberg
        - https://www.britannica.com/science/Konigsberg-bridge-problem

- (basically can you start at some point of the bridge, go through each of the 7 bridges and do that without repeating any bridges)
- 

→ Possible Presentation Ideas

1. Website (prob the easiest)
2. Video